THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE · NCERT-TYPE UNIT TEST SET 1

Chapters 5–8: **Describing Motion Around Us, How Forces Affect Motion, Work, Energy and Simple Machines, Sound Waves: Characteristics and Applications** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. NCERT-type: questions in the style of the textbook examples and exercises.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. A runner completes one full lap of a 400 m circular track. Her displacement at the end is: 1

(a) 0 **(b)** 200 m **(c)** 800 m **(d)** 400 m

2. A body continues in its state of rest or of uniform motion in a straight line unless acted on by an external force. This is: 1

(a) Newton's second law **(b)** Newton's third law
(c) the law of gravitation **(d)** Newton's first law

3. In science, work is done when: 1

(a) a force acts but there is no displacement
(b) a force acts on a body and the body is displaced in the direction of the force
(c) a person holds a heavy load at rest
(d) a person thinks hard

4. Sound is produced by: 1

(a) vibrating bodies **(b)** heat only
(c) stationary bodies **(d)** light

5. The slope of a velocity-time graph gives: 1

(a) acceleration **(b)** speed **(c)** force **(d)** distance

6. According to Newton's third law, action and reaction forces: 1

(a) act on the same body
(b) are unequal
(c) are equal in magnitude, opposite in direction and act on different bodies
(d) cancel each other on the same body

7. The work-energy theorem states that the work done on a body is equal to the change in its: 1

(a) kinetic energy **(b)** mass **(c)** momentum **(d)** potential energy

8. The loudness of a sound depends mainly on its: 1

(a) amplitude **(b)** speed **(c)** wavelength **(d)** frequency

9. The equation v squared = u squared + 2as relates: 1

(a) velocity, acceleration and displacement **(b)** velocity and time only
(c) work and time **(d)** force and mass

10. The weight of a body on the Moon compared to that on Earth is about: 1

(a) one-sixth **(b)** zero **(c)** the same **(d)** six times

11. A simple machine is a device that: 1

(a) makes work easier by changing the magnitude or direction of force
(b) creates energy
(c) needs no effort
(d) does more work than the input

12. The minimum distance of the reflecting surface to hear an echo in air (v = 344 m/s) is about: 1

(a) 1.72 m **(b)** 344 m **(c)** 34.4 m **(d)** 17.2 m

13. The distance travelled by a body is always: 1

(a) equal to velocity
(b) less than displacement
(c) negative
(d) greater than or equal to the magnitude of displacement

14. Inertia of a body depends on its: 1

(a) colour **(b)** speed only **(c)** volume **(d)** mass

15. The SI unit of work is the: 1

(a) watt **(b)** newton **(c)** joule **(d)** pascal

SECTION — B (Very Short Answer Type) — 5 × 2 = 10

Answer in about 20–30 words each.

16. Differentiate between distance and displacement. 2

17. State Newton's first law of motion. What is inertia? 2

18. Define work in the scientific sense. Give its SI unit. 2

19. How is sound produced? Name the medium needed for its propagation. 2

20. Define acceleration. Give its SI unit. 2

SECTION — C (Short Answer Type) — 5 × 3 = 15

Answer in about 40–60 words each.

21. A boy walks 3 m east and then 4 m north. Find the distance and the magnitude of displacement. 3

22. A body of mass 2 kg is acted on by forces of 8 N east and 3 N west. Find the net force and acceleration. 3

23. A girl lifts a dumbbell of 5 kg to a height of 1.2 m and then lowers it. State when positive and when negative work is done and calculate the work done by her force in lifting (g = 10). 3

24. A wave has a frequency of 200 Hz and a wavelength of 1.7 m. Find its speed and time period. 3

OR

From a velocity-time graph of a body moving at 6 m/s for 10 s find the distance covered. 3

25. A force of 12 N acts for 3 s on a body at rest of mass 4 kg. Find the velocity acquired. 3

OR

Calculate the energy consumed and the cost if a 100 W bulb is used for 10 hours and the rate is 6 rupees per unit. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Derive the three equations of motion for a body moving with uniform acceleration using the velocity-time graph or algebra. 5

OR

State Newton's three laws of motion and give one example of each. 5

27. Explain the work done by a constant force, with the cases of force along, opposite and perpendicular to the displacement. Calculate the work done by a force of 15 N moving a body 4 m along the force. 5

OR

Explain the production and propagation of sound waves, with the definitions of compression, rarefaction, amplitude, wavelength, frequency and time period. 5

END OF QUESTION PAPER Total: 50 Marks

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